

Task 2: Retrieval Optimization & Evaluation



Overview

In this task, your objective is to optimize, experiment with, and quantitatively evaluate the retrieval stage of the RAG pipeline using the Jupyter notebook:

- `Task2_Retrieval_Optimization.ipynb`

While Task 1 focused on ingesting documents and building the vector index, Task 2 shifts focus to **retrieval quality and evaluation** — systematically proving that the vector database retrieves the *right* evidence using empirical benchmarks rather than subjective impressions.



Objectives

By completing this task, you will:

1. **Analyze `top_k` Trade-offs:** Understand the balance between retrieval precision (avoiding noise/drift) and recall/coverage (not missing evidence).
2. **Conduct Chunking Ablation Experiments:** Perform controlled experiments comparing small, balanced, and large chunk configurations (`chunk_size` and `chunk_overlap`).
3. **Benchmarking with Evaluation Test Sets:** Load real clinical test queries and verified expected sources to evaluate retrieval performance.
4. **Calculate Quantitative Metrics (Precision@k):** Implement and compute page-level Retrieval Precision@k across the test benchmark.
5. **Data-Driven Configuration:** Select and justify the optimal chunking and retrieval parameters based on empirical results.



Step-by-Step Task Guide

1. Setup & Index Verification

- Open `Task2_Retrieval_Optimization.ipynb`.
- Run the setup cell to rebuild/load the index and confirm the chunk count and metadata integrity.

2. Evaluate `top_k` Behavior & Context Drift

- Run retrieval on sample clinical queries across multiple k values ($k = 1, k = 3, k = 8$).
- Observe how increasing k broadens context but risks introducing irrelevant guideline sections (noise drift).

3. Run Chunking Ablation Experiments

- Execute the ablation experiment evaluating three distinct configurations:
 - **Small:** `chunk_size = 200, chunk_overlap = 0`
 - **Balanced:** `chunk_size = 400, chunk_overlap = 50`
 - **Large:** `chunk_size = 600, chunk_overlap = 100`
- Compare chunk volume against average top-3 relevance scores to identify the optimal configuration.

4. Load the Evaluation Benchmark

- Ingest test questions and expected citation sources from the evaluation dataset.
- Inspect test question formats, clinical topics, and expected document/page mappings.

5. Calculate Retrieval Precision@k

- Compute page-level **Precision@k** ($\text{Precision@k} = \frac{\text{relevant chunks in top-}k}{k}$) across the test queries.
- Correctly handle out-of-scope control questions to verify system selectivity.
- Compute the overall Average Precision@k score across all scored questions.

Acceptance Criteria (Definition of Done)

- ☐ All cells in `Task2_Retrieval_Optimization.ipynb` execute sequentially from top to bottom without errors.
- ☐ `top_k` comparison is executed across $k = [1, 3, 8]$ with observed trade-offs documented.
- ☐ Chunking ablation experiment runs across all 3 configurations and produces average relevance scores.
- ☐ Precision@k metric is calculated for the benchmark test set with average precision reported.
- ☐ The best-performing chunking and retrieval configuration is identified with empirical data backing.